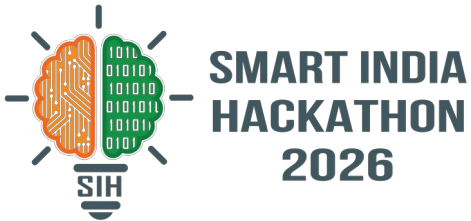


SMART INDIA HACKATHON 2026



RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

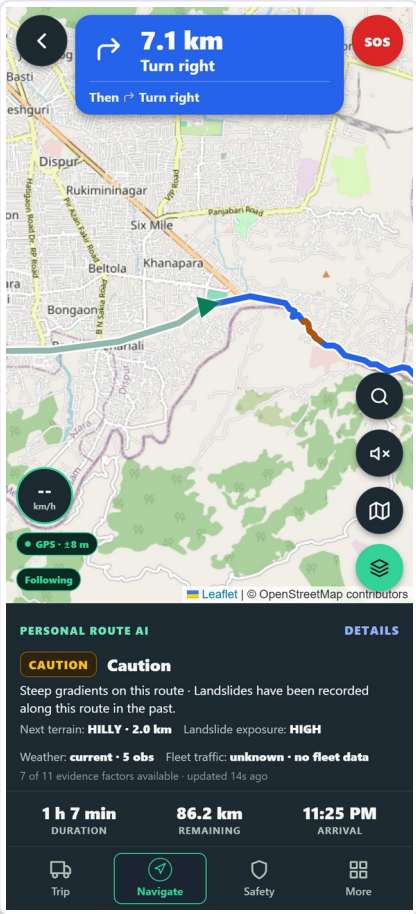
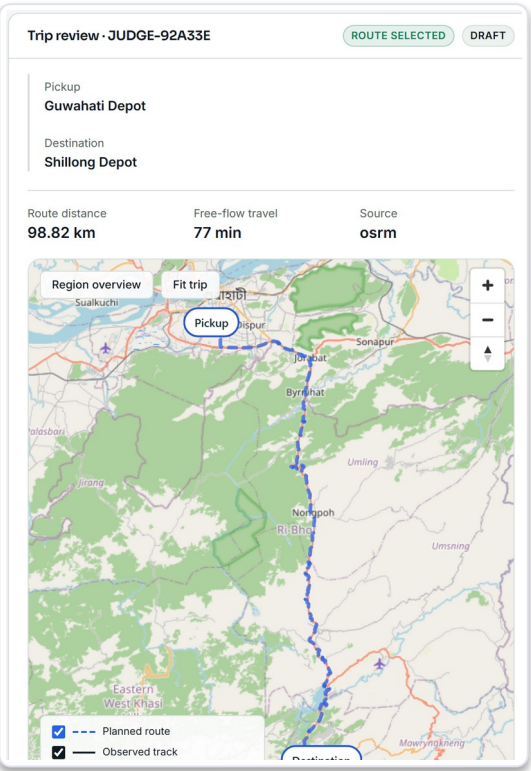
SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

SIH26002	
PROBLEM STATEMENT ID	MDoNER • Smart Automation • Software
PROBLEM STATEMENT TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
THEME	Smart Automation
PS CATEGORY	Software
TEAM ID	17 (internal group no.)
TEAM NAME	NER-AI LOGISTICS

End-to-end working prototype • physical-phone validated • live demo corridor Guwahati → Shillong

LIVE WORKING SYSTEM



Manager route review • Driver navigation — real screens

11

ROUTE FACTORS

scored on the selected road before a truck is dispatched, each with its own freshness

A normal router checks none of them.

Origin → Shortest / fastest road → Truck

No governance. No answer when data is missing.

Not the shortest road. The road that is usable now.



UNKNOWN ≠ SAFE

Missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

MANAGER • CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN
0.5 KM STEEP

49-1436 m · 1975 m climb · steepest 10.3%

flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km

Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY
HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km

6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two

NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS
NORMAL

Discharge at 5 river cells along the corridor, highest 1.0x its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS
NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC
UNKNOWN

Real screen · evidence · freshness · UNKNOWN

60 s

ROUTE-AHEAD MONITOR

the road ahead is re-scored every minute while the truck is moving; a material change goes back through the policy

A reroute is proposed to the manager, never applied silently.

TECHNICAL APPROACH

VERIFIED DATA**WEATHER**Open-Meteo · MET
Norway**TERRAIN**Copernicus DEM ·
OpenTopoData**WARNINGS**NDMA SACHET
(CAP)**FLOOD**

GloFAS

LANDSLIDE

NASA GLC 2007-17

TRAFFIC

fleet GPS fixes

ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE

CAUTION

HOLD

REROUTE

MANAGER REVIEW

DRIVER NAVIGATION

60 s WATCH

REASSESS / REROUTE

CLIENTS Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

WHO DECIDES**DETERMINISTIC SAFETY ENGINE**

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN

Verified evidence



Deterministic safety policy



Manager governance



Driver action

FEASIBILITY AND VIABILITY

12/12

JUDGE-FLOW STEPS

dispatch → accept → truck check → GPS
navigation → real off-route → reroute → manager
approval → stops → delivery, on a physical
Android phone

9/9 role-flow steps

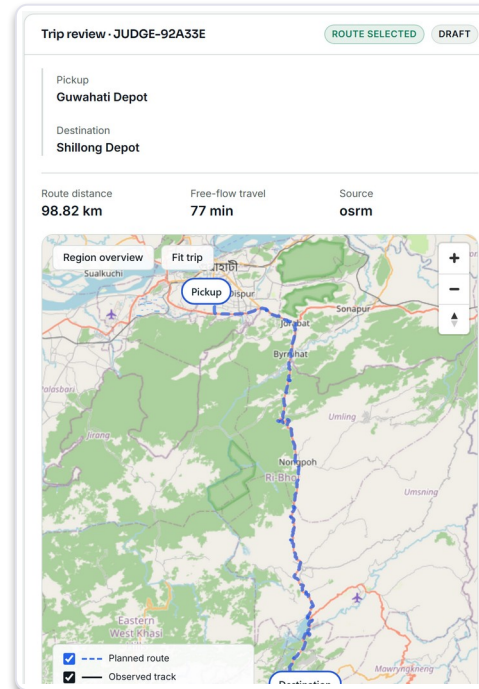
1138+ backend · 621+ driver · 170+ manager
tests

PHYSICAL ANDROID · CERTIFIED

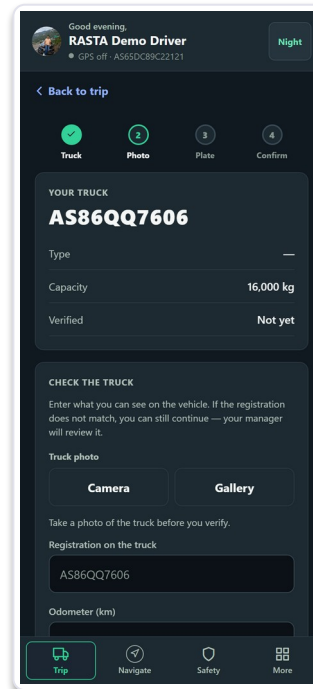
Canonical demo

Guwahati → Shillong · ≈ 98.8 km real route

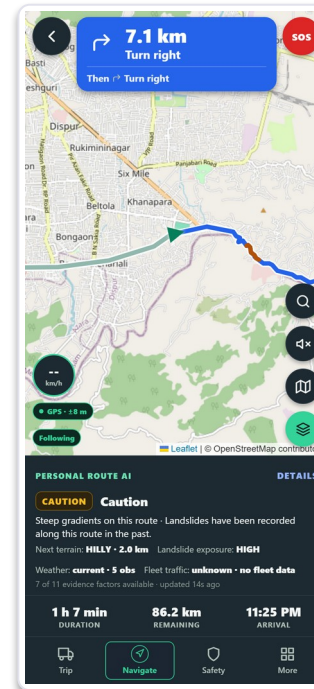
Not a concept. A working end-to-end prototype.



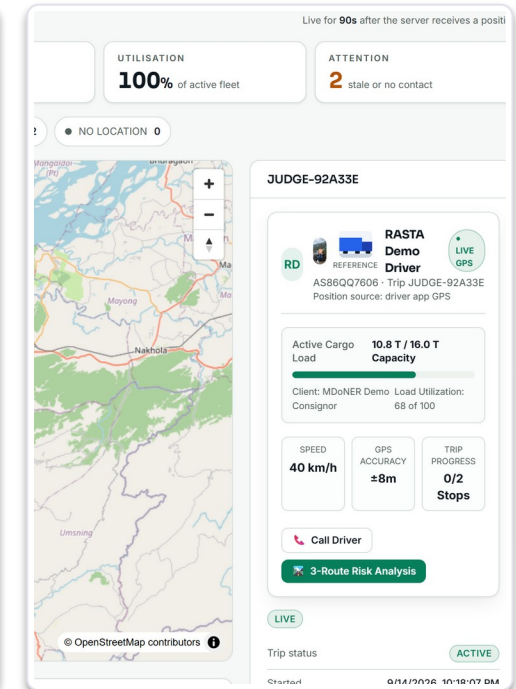
Manager · route review



Driver · truck check



Driver · navigation



Manager · reroute review

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

IMPACT AND BENEFITS

0

UNKNOWN FACTORS COUNTED AS SAFE

the rule the whole product is built on: missing or stale evidence is never read as a clear road

FOOD • MEDICINE • FUEL • RELIEF SUPPLIES

essential logistics for hill communities, decided on evidence

RASTA AI does not ask only “Which road is shortest?” It asks
“Can this truck reliably use this corridor now?”

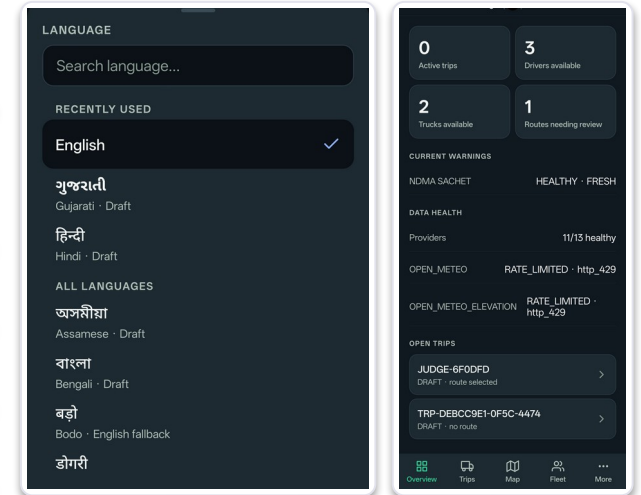
WHO BENEFITS

FLEET MANAGERS	Route usability and uncertainty visible before dispatch; auditable decisions.
DRIVERS	Safer navigation, danger context, emergency support, 23 languages.
REMOTE COMMUNITIES	More resilient access to food, medicine, fuel and relief supplies.
GOVERNMENT / OPERATIONS	Auditable route evidence and disruption visibility per corridor.

DEMONSTRATED TODAY

23	selectable languages, explicit fallback status	12/12	judge-flow steps certified on a physical phone
11	route factors in every decision	100 %	of reroutes reviewed by a human
60 s	route-ahead monitor while driving		

REAL SCREENS • APK 1.0.18



Driver • 23 languages

Manager • mobile

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

HUMAN GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY



95 %

RECALL

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE

NOT DEPLOYED

FPR 50 %

too high for production deployment

VALIDATION BEFORE AUTOMATION.

Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM • OpenTopoData

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

LANDSLIDE HISTORY

NASA Global Landslide Catalog

historical exposure; inventory 2007–2017, ≤5 km location-accuracy filter

AI ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

PROBLEM STATEMENT

MDoNER • SIH26002

AI-based smart logistics and accessibility intelligence for the NER

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 • calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope • docs/HAZARD_ERROR_ANALYSIS.md

REFERENCES gpm.nasa.gov/landslides (NASA GLC) • sachet.ndma.gov.in • global-flood.emergency.copernicus.eu (GloFAS) • open-meteo.com • api.met.no • opentopodata.org • Copernicus DEM • project-osrm.org • openstreetmap.org • github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry)